



Programme: F.E. (All Branches)

Internal Assessment Test–I (Sample) (A.Y. 2026–27)

NEP – 2020

Subject: Applied Physics (BSC102)

Semester: I

Max. Marks: 15

Date: 19/10/2026

Time: 3:00 to 4:00

Q. No.	Questions	Marks	BL	CO
Q.1	Attempt any two of the following:			
1(a)	Why does a soap film show different colours when seen in white light? Explain the reason with the help of a suitable diagram.	3	L2	CO3
1(b)	In a Newton's rings experiment, the diameter of the 10th dark ring is 0.47 cm, and that of the 5th dark ring is 0.27 cm. The radius of curvature of the plano-convex lens used is 100 cm. Calculate the wavelength of the light used in the experiment.	3	L3	CO3
1(c)	Illustrate the conditions for constructive and destructive interference in thin films for reflected and transmitted waves. Discuss the effect of phase change on reflection with a labelled diagram.	3	L2	CO3
Q.2	Attempt any one of the following:			
2(a)	Obtain the formula for the Numerical Aperture (NA) of a step-index optical fiber.	4	L3	CO2
2(b)	i) Compare step-index and graded-index optical fibers based on their refractive index profiles and working characteristics. ii) Find the Numerical Aperture (NA) and the acceptance angle of an optical fiber if the refractive index of the core is 1.57 and that of the cladding is 1.55. Assume the fiber is surrounded by air.	4	L3	CO2
Q.3	Attempt any one of the following:			
3(a)	With the help of a labelled energy level diagram, explain the construction and working of He-Ne Laser. Describe the role of Resonator Cavity.	5	L2, L3	CO1
3(b)	i) A light source has a spectral bandwidth of 5×10^{12} Hz. Calculate its coherence time and coherence length. ii) Explain the terms; Metastable state, Population inversion, Spontaneous emission, Stimulated emission.	5	L2, L3	CO1

BL: Bloom's Taxonomy Levels

(L1: Remembering, L2: Understanding, L3: Applying, L4: Analyzing, L5: Evaluating, L6: Creating)

CO: Course Outcomes

	Solutions	
1(a)	Why does a soap film appear colorful when viewed in white light? Explain with suitable diagram.	3 Marks
	Explanation of phenomena. Labelled diagram.	1.5 1.5
1(b)	In a Newton's rings experiment, the diameter of the 10th dark ring is 0.47 cm, and that of the 5th dark ring is 0.27 cm. The radius of curvature of the plano-convex lens used is 100 cm. Calculate the wavelength of the light used in the experiment.	3 Marks
	Answer: Given: Diameter of 10th dark ring, $D_{10} = 0.47$ cm Diameter of 5th dark ring, $D_5 = 0.27$ cm Radius of curvature, $R = 100$ cm Formula: For Newton's rings, $D_n^2 = 4nR\lambda$ Therefore, $D_{10}^2 - D_5^2 = 4R\lambda(10 - 5)$ Substituting the values: $(0.47)^2 - (0.27)^2 = 4 \times 100 \times \lambda \times 5$ $0.2209 - 0.0729 = 2000\lambda$ $0.1480 = 2000\lambda$ $\lambda = \frac{0.1480}{2000}$ $\lambda = 7.4 \times 10^{-5} \text{ cm} \quad \text{or} \quad \lambda = 740 \text{ nm}$	1.5 1.5
1(c)	Illustrate the conditions for constructive and destructive interference in thin films for reflected and transmitted waves. Discuss the effect of phase change on reflection with a labelled diagram.	3 Marks
	Diagram for interference in thin film. Conditions for maxima & minima.	1 2
2(a)	Obtain the formula for the Numerical Aperture (NA) of a step-index optical fiber.	4 Marks
	Diagram of wave propagation Derivation of NA	1 3
2(b)	(i) Compare step-index and graded-index optical fibers based on their refractive index profiles and working characteristics.	2 Marks
	Any five points of differentiation.	2
	(ii) Find the Numerical Aperture (NA) and the acceptance angle of an optical fiber if the refractive index of the core is 1.57 and cladding is 1.55. Assume the fiber is surrounded by air.	2 Marks
	Refractive index of core, $n_1 = 1.57$ Refractive index of cladding, $n_2 = 1.55$ Refractive index of air, $n_0 = 1$	1 1

	Numerical Aperture (NA) The Numerical Aperture is given by: $NA = \sqrt{n_1^2 - n_2^2}$ Substituting the values: $NA = \sqrt{(1.57)^2 - (1.55)^2}$ $NA = \sqrt{2.4649 - 2.4025}$ $NA = \sqrt{0.0624}$ $NA = 0.2498 \implies NA \approx 0.25$ Acceptance Angle The acceptance angle is given by: $NA = n_0 \sin \theta_a$ Since the fiber is surrounded by air, $n_0 = 1$. Therefore, $\sin \theta_a = \frac{NA}{n_0} = \frac{0.2498}{1}$ $\theta_a = \sin^{-1}(0.2498) \approx 14.47^\circ$ Final Answer: Numerical Aperture (NA) ≈ 0.25 Acceptance angle (θ_a) $\approx 14.47^\circ$	1 1
3(a)	With the help of a labelled energy level diagram, explain the construction and working of He-Ne Laser. Describe the role of Resonator Cavity.	5 Marks
	Energy level diagram Construction & Working Role of Resonator Cavity.	1 3 1
3(b)	i) A light source has a spectral bandwidth of 5×10^{12} Hz. Calculate its coherence time and coherence length.	2.5 Marks
	Spectral bandwidth, $\Delta\nu = 5 \times 10^{12}$ Hz Speed of light, $c = 3 \times 10^8$ m/s Coherence Time The coherence time is given by: $\tau_c = \frac{1}{\Delta\nu} = \frac{1}{5 \times 10^{12}} = 2 \times 10^{-13} \text{ s}$ Hence, the coherence time is $\tau_c = 2 \times 10^{-13}$ s. Coherence Length The coherence length is given by: $L_c = c \cdot \tau_c = (3 \times 10^8) \cdot (2 \times 10^{-13}) = 6 \times 10^{-5} \text{ m}$ Hence, the coherence length is $L_c = 6 \times 10^{-5} \text{ m} = 60 \mu\text{m}$.	2 0.5
	ii) Explain the terms; Metastable state, Population inversion, Spontaneous emission, Stimulated emission.	2.5 Marks
	Explanation of various terms.	2.5